



End-to-End Testing

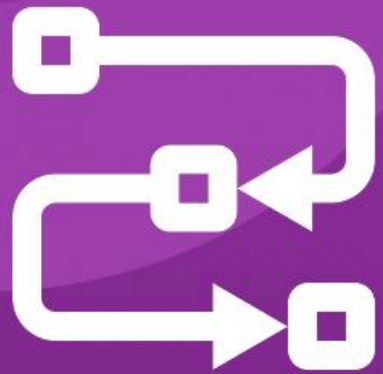
Testing to validate software workflows for the entire system

In the previous video...

Integration Testing



In this video...



E2E Testing

What is E2E Testing?

- A simulation of user interaction with the system
- Done to ensure that all integrated components function correctly
- Covers the entire application stack that includes database interactions, network requests, and API's
- Validates the system under production-like conditions
- Catches bugs that other tests miss and ensures system cohesion

How does it differ from integration testing?



Integration testing



End-to-end testing

Framework



Playwright

JS

Testing for correct result

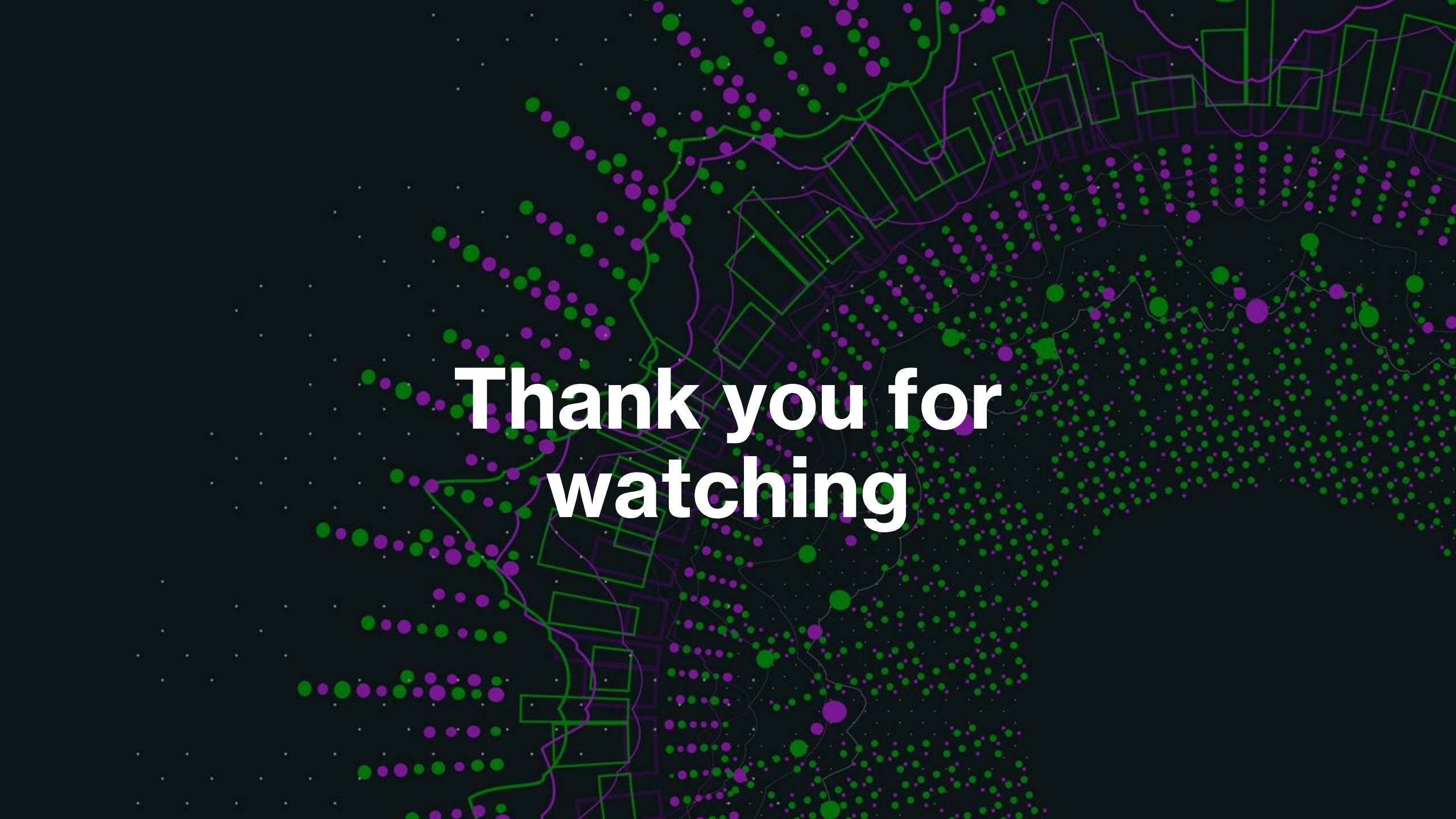


```
test('should calculate 2 + 3', async ({page}) => {  
  await page.goto('http://localhost:5173'); //Link to the frontend  
  
  // On the frontend code, give each of the components this attribute: data-testid=""  
  
  await page.getByTestId('btn-2').click(); // Button for 2  
  await page.getByTestId('btn-plus').click(); // Button for plus  
  await page.getByTestId('btn-3').click(); // Button for 3  
  
  await page.getByTestId('btn-equals').click(); // button for equals  
  
  await expect(  
    page.getByTestId('calculator-output')  
  ).toHaveText('5'); // Expected result of '5'  
});
```

Testing for error result



```
test('should return and display an error for empty input', async ({page}) => {  
  await page.goto('http://localhost:5173'); // Link to the frontend  
  
  await page.getByTestId('btn-equals').click(); // Button for equals  
  
  page.on('dialog', async (dialog) => { // Alert message that contains error message  
    expect(dialog.message()).toContain( // What the error should contain  
      'Input cannot be empty'  
    );  
  
    await dialog.accept(); // Accept dialog  
  });  
  
  await expect(  
    page.getByTestId('calculator-output')  
  ).toHaveText('Error'); // Expected output  
});
```


The background is a black field filled with a complex pattern of green and purple dots, lines, and rectangles. The dots are arranged in various clusters and trails. The lines are thin and wavy, connecting some of the dots. The rectangles are thin and appear to be floating or overlapping the other elements. The overall effect is a dense, abstract, and somewhat chaotic visual field.

**Thank you for
watching**